

f has the desired properties, and the proof of Weierstrass's theorem is complete.

5 Hadamard's factorization theorem

The theorem of this section combines the results relating the growth of a function to the number of zeros it possesses, and the above product theorem. Weierstrass's theorem states that a function that vanishes at the points $a_1, a_2, \dots$ takes the form

$$e^{g(z)} z^m \prod_{n=1}^{\infty} E_n(z/a_n).$$

Hadamard refined this result by showing that in the case of functions of finite order, the degree of the canonical factors can be taken to be constant, and g is then a polynomial.

Recall that an entire function has an order of growth $\leq \rho$ if

$$|f(z)| \leq A e^{B|z|^\rho},$$

and that *the* order of growth ρ_0 of f is the infimum of all such ρ 's.

A basic result we proved earlier was that if f has order of growth $\leq \rho$, then

$$n(r) \leq C r^\rho, \quad \text{for all large } r,$$

and if $a_1, a_2, \dots$ are the non-zero zeros of f , and $s > \rho$, then

$$\sum |a_n|^{-s} < \infty.$$

Theorem 5.1 *Suppose f is entire and has growth order ρ_0 . Let k be the integer so that $k \leq \rho_0 < k + 1$. If $a_1, a_2, \dots$ denote the (non-zero) zeros of f , then*

$$f(z) = e^{P(z)} z^m \prod_{n=1}^{\infty} E_k(z/a_n),$$

where P is a polynomial of degree $\leq k$, and m is the order of the zero of f at $z = 0$.

Main lemmas

Here we gather a few lemmas needed in the proof of Hadamard's theorem.

Lemma 5.2 *The canonical products satisfy*

$$|E_k(z)| \geq e^{-c|z|^{k+1}} \quad \text{if } |z| \leq 1/2$$

and

$$|E_k(z)| \geq |1 - z| e^{-c'|z|^k} \quad \text{if } |z| \geq 1/2.$$

Proof. If $|z| \leq 1/2$ we can use the power series to define the logarithm of $1 - z$, so that

$$E_k(z) = e^{\log(1-z) + \sum_{n=1}^k z^n/n} = e^{-\sum_{n=k+1}^{\infty} z^n/n} = e^w.$$

Since $|e^w| \geq e^{-|w|}$ and $|w| \leq c|z|^{k+1}$, the first part of the lemma follows. For the second part, simply observe that if $|z| \geq 1/2$, then

$$|E_k(z)| = |1 - z| |e^{z+z^2/2+\dots+z^k/k}|,$$

and that there exists $c' > 0$ such that

$$|e^{z+z^2/2+\dots+z^k/k}| \geq e^{-|z+z^2/2+\dots+z^k/k|} \geq e^{-c'|z|^k}.$$

The inequality in the lemma when $|z| \geq 1/2$ then follows from these observations.

The key to the proof of Hadamard's theorem consists of finding a lower bound for the product of the canonical factors when z stays away from the zeros $\{a_n\}$. Therefore, we shall first estimate the product from below, in the complement of small discs centered at these points.

Lemma 5.3 *For any s with $\rho_0 < s < k + 1$, we have*

$$\left| \prod_{n=1}^{\infty} E_k(z/a_n) \right| \geq e^{-c|z|^s},$$

except possibly when z belongs to the union of the discs centered at a_n of radius $|a_n|^{-k-1}$, for $n = 1, 2, 3, \dots$

Proof. The proof this lemma is a little subtle. First, we write

$$\prod_{n=1}^{\infty} E_k(z/a_n) = \prod_{|a_n| \leq 2|z|} E_k(z/a_n) \prod_{|a_n| > 2|z|} E_k(z/a_n).$$

For the second product the estimate asserted above holds with no restriction on z . Indeed, by the previous lemma

$$\begin{aligned} \left| \prod_{|a_n| > 2|z|} E_k(z/a_n) \right| &= \prod_{|a_n| > 2|z|} |E_k(z/a_n)| \\ &\geq \prod_{|a_n| > 2|z|} e^{-c|z/a_n|^{k+1}} \\ &\geq e^{-c|z|^{k+1} \sum_{|a_n| > 2|z|} |a_n|^{-k-1}}. \end{aligned}$$

But $|a_n| > 2|z|$ and $s < k+1$, so we must have

$$|a_n|^{-k-1} = |a_n|^{-s} |a_n|^{s-k-1} \leq C |a_n|^{-s} |z|^{s-k-1}.$$

Therefore, the fact that $\sum |a_n|^{-s}$ converges implies that

$$\left| \prod_{|a_n| > 2|z|} E_k(z/a_n) \right| \geq e^{-c|z|^s}$$

for some $c > 0$.

To estimate the first product, we use the second part of Lemma 5.2, and write

$$(5) \quad \left| \prod_{|a_n| \leq 2|z|} E_k(z/a_n) \right| \geq \prod_{|a_n| \leq 2|z|} \left| 1 - \frac{z}{a_n} \right| \prod_{|a_n| \leq 2|z|} e^{-c'|z/a_n|^k}.$$

We now note that

$$\prod_{|a_n| \leq 2|z|} e^{-c'|z/a_n|^k} = e^{-c'|z|^k \sum_{|a_n| \leq 2|z|} |a_n|^{-k}},$$

and again, we have $|a_n|^{-k} = |a_n|^{-s} |a_n|^{s-k} \leq C |a_n|^{-s} |z|^{s-k}$, thereby proving that

$$\prod_{|a_n| \leq 2|z|} e^{-c'|z/a_n|^k} \geq e^{-c|z|^s}.$$

It is the estimate on the first product on the right-hand side of (5) which requires the restriction on z imposed in the statement of the

lemma. Indeed, whenever z does not belong to a disc of radius $|a_n|^{-k-1}$ centered at a_n , we must have $|a_n - z| \geq |a_n|^{-k-1}$. Therefore

$$\begin{aligned} \prod_{|a_n| \leq 2|z|} \left| 1 - \frac{z}{a_n} \right| &= \prod_{|a_n| \leq 2|z|} \left| \frac{a_n - z}{a_n} \right| \\ &\geq \prod_{|a_n| \leq 2|z|} |a_n|^{-k-1} |a_n|^{-1} \\ &= \prod_{|a_n| \leq 2|z|} |a_n|^{-k-2}. \end{aligned}$$

Finally, the estimate for the first product follows from the fact that

$$\begin{aligned} (k+2) \sum_{|a_n| \leq 2|z|} \log |a_n| &\leq (k+2) \mathbf{n}(2|z|) \log 2|z| \\ &\leq c|z|^s \log 2|z| \\ &\leq c'|z|^{s'}, \end{aligned}$$

for any $s' > s$, and the second inequality follows because $\mathbf{n}(2|z|) \leq c|z|^s$ by Theorem 2.1. Since we restricted s to satisfy $s > \rho_0$, we can take an initial s sufficiently close to ρ_0 , so that the assertion of the lemma is established (with s being replaced by s').

Corollary 5.4 *There exists a sequence of radii, $r_1, r_2, \dots$, with $r_m \rightarrow \infty$, such that*

$$\left| \prod_{n=1}^{\infty} E_k(z/a_n) \right| \geq e^{-c|z|^s} \quad \text{for } |z| = r_m.$$

Proof. Since $\sum |a_n|^{-k-1} < \infty$, there exists an integer N so that

$$\sum_{n=N}^{\infty} |a_n|^{-k-1} < 1/10.$$

Therefore, given any two consecutive large integers L and $L+1$, we can find a positive number r with $L \leq r \leq L+1$, such that the circle of radius r centered at the origin does not intersect the forbidden discs of Lemma 5.3. For otherwise, the union of the intervals

$$I_n = \left[|a_n| - \frac{1}{|a_n|^{k+1}}, |a_n| + \frac{1}{|a_n|^{k+1}} \right]$$

(which are of length $2|a_n|^{-k-1}$) would cover all the interval $[L, L+1]$. (See Figure 1.) This would imply $2 \sum_{n=N}^{\infty} |a_n|^{-k-1} \geq 1$, which is a contradiction. We can then apply the previous lemma with $|z| = r$ to conclude the proof of the corollary.

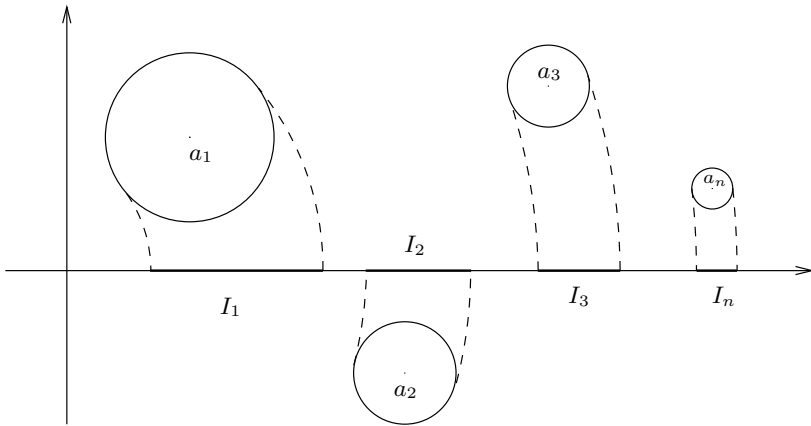


Figure 1. The intervals I_n

Proof of Hadamard's theorem

Let

$$E(z) = z^m \prod_{n=1}^{\infty} E_k(z/a_n).$$

To prove that E is entire, we repeat the argument in the proof of Theorem 4.1; we take into account that by Lemma 4.2

$$|1 - E_k(z/a_n)| \leq c \left| \frac{z}{a_n} \right|^{k+1} \quad \text{for all large } n,$$

and that the series $\sum |a_n|^{-k-1}$ converges. (Recall $\rho_0 < s < k+1$.) Moreover, E has the zeros of f , therefore f/E is holomorphic and nowhere vanishing. Hence

$$\frac{f(z)}{E(z)} = e^{g(z)}$$

for some entire function g . By the fact that f has growth order ρ_0 , and because of the estimate from below for E obtained in Corollary 5.4, we have

$$e^{\operatorname{Re}(g(z))} = \left| \frac{f(z)}{E(z)} \right| \leq c' e^{c|z|^s}$$

for $|z| = r_m$. This proves that

$$\operatorname{Re}(g(z)) \leq C|z|^s, \quad \text{for } |z| = r_m.$$

The proof of Hadamard's theorem is therefore complete if we can establish the following final lemma.

Lemma 5.5 *Suppose g is entire and $u = \operatorname{Re}(g)$ satisfies*

$$u(z) \leq Cr^s \quad \text{whenever } |z| = r$$

for a sequence of positive real numbers r that tends to infinity. Then g is a polynomial of degree $\leq s$.

Proof. We can expand g in a power series centered at the origin

$$g(z) = \sum_{n=0}^{\infty} a_n z^n.$$

We have already proved in the last section of Chapter 3 (as a simple application of Cauchy's integral formulas) that

$$(6) \quad \frac{1}{2\pi} \int_0^{2\pi} g(re^{i\theta}) e^{-in\theta} d\theta = \begin{cases} a_n r^n & \text{if } n \geq 0 \\ 0 & \text{if } n < 0. \end{cases}$$

By taking complex conjugates we find that

$$(7) \quad \frac{1}{2\pi} \int_0^{2\pi} \overline{g(re^{i\theta})} e^{-in\theta} d\theta = 0$$

whenever $n > 0$, and since $2u = g + \overline{g}$ we add equations (6) and (7) to obtain

$$a_n r^n = \frac{1}{\pi} \int_0^{2\pi} u(re^{i\theta}) e^{-in\theta} d\theta, \quad \text{whenever } n > 0.$$

For $n = 0$ we can simply take real parts of both sides of (6) to find that

$$2\operatorname{Re}(a_0) = \frac{1}{\pi} \int_0^{2\pi} u(re^{i\theta}) d\theta.$$

Now we recall the simple fact that whenever $n \neq 0$, the integral of $e^{-in\theta}$ over any circle centered at the origin vanishes. Therefore

$$a_n = \frac{1}{\pi r^n} \int_0^{2\pi} [u(re^{i\theta}) - Cr^s] e^{-in\theta} d\theta \quad \text{when } n > 0,$$

hence

$$|a_n| \leq \frac{1}{\pi r^n} \int_0^{2\pi} [Cr^s - u(re^{i\theta})] d\theta \leq 2Cr^{s-n} - 2\operatorname{Re}(a_0)r^{-n}.$$

Letting r tend to infinity along the sequence given in the hypothesis of the lemma proves that $a_n = 0$ for $n > s$. This completes the proof of the lemma and of Hadamard's theorem.

6 Exercises

1. Give another proof of Jensen's formula in the unit disc using the functions (called Blaschke factors)

$$\psi_\alpha(z) = \frac{\alpha - z}{1 - \bar{\alpha}z}.$$

[Hint: The function $f/(\psi_{z_1} \cdots \psi_{z_N})$ is nowhere vanishing.]

2. Find the order of growth of the following entire functions:

- (a) $p(z)$ where p is a polynomial.
- (b) e^{bz^n} for $b \neq 0$.
- (c) e^{e^z} .

3. Show that if τ is fixed with $\operatorname{Im}(\tau) > 0$, then the Jacobi theta function

$$\Theta(z|\tau) = \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{2\pi i n z}$$

is of order 2 as a function of z . Further properties of Θ will be studied in Chapter 10.

[Hint: $-n^2 t + 2n|z| \leq -n^2 t/2$ when $t > 0$ and $n \geq 4|z|/t$.]

4. Let $t > 0$ be given and fixed, and define $F(z)$ by

$$F(z) = \prod_{n=1}^{\infty} (1 - e^{-2\pi n t} e^{2\pi i z}).$$

Note that the product defines an entire function of z .